## Experiment 16: Implementation of Feedforward Neural Network Using Python

**Aim:**

To implement a Feedforward Neural Network using Python to process input data and generate output using an activation function.

**Software Required:**

- Python 3.x
- NumPy library

**Theory:**

A Feedforward Neural Network is an artificial neural network in which information flows in only one direction, from the input layer to the hidden layer and then to the output layer. It consists of neurons, weights, biases, and activation functions. The sigmoid activation function is used to produce output values between 0 and 1.

**Algorithm:**

1. Import the NumPy library.
2. Define the input data and target output.
3. Initialize weights and biases.
4. Define the sigmoid activation function.
5. Calculate the hidden layer output.
6. Calculate the final output using the output layer.
7. Display the predicted output.

**Program:**

```
import numpy as np

# Input data

X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])

# Target output (AND gate)

y = np.array([[0], [0], [0], [1]])

# Initialize weights and bias

np.random.seed(1)

W1 = np.random.rand(2, 3)
```

```
W2 = np.random.rand(3, 1)

b1 = np.zeros((1, 3))

b2 = np.zeros((1, 1))

# Sigmoid activation function

def sigmoid(x):

    return 1 / (1 + np.exp(-x))

# Feedforward process

hidden = sigmoid(np.dot(X, W1) + b1)

output = sigmoid(np.dot(hidden, W2) + b2)

# Display output

print("Input:")

print(X)

print("Predicted Output:")

print(output)
```

exp 16 feed forward.py - C:/Users/yadal/OneDrive/Documents/AI/exp 16 feed forward.py (3....

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```
import numpy as np

# Input data
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# Target output (AND gate)
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np.random.seed(1)
W1 = np.random.rand(2, 3)
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# Sigmoid activation function
def sigmoid(x):
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# Feedforward process
hidden = sigmoid(np.dot(X, W1) + b1)
output = sigmoid(np.dot(hidden, W2) + b2)

# Display output
print("Input:")
print(X)
print("Predicted Output:")
print(output)
```

**Output:**

IDLE Shell 3.14.7

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```
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 16 feed forward.py =====
Input:
[[0 0]
 [0 1]
 [1 0]
 [1 1]]
Predicted Output:
[[0.61403239]
 [0.62247681]
 [0.63253756]
 [0.64017496]]
>>>
```

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**Result:**

Thus, the Feedforward Neural Network was implemented successfully using Python and NumPy.